MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE LDN

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

*2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

*2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787, except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.22 failures per million operating hours for class I or 0.22 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be LDN.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/19-1 for class I and M28787/19-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (I_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4 QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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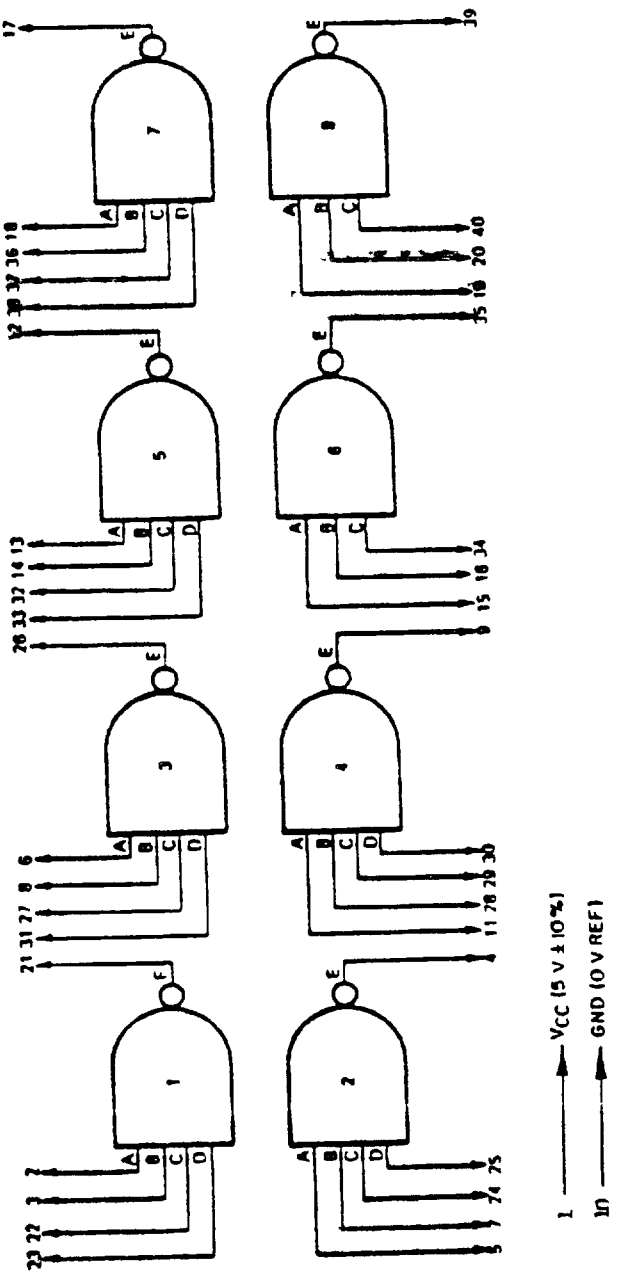


FIGURE 1. LDN module function diagram, NAND gate

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*TABLE I. Module electrical requirements

Test condition	Parameter	Ref test table	Test type 1/ 3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A8	V_{OUT} (low loaded)	III	S	0.01	0.80	0.01	0.80	0.01	0.80	0.01	0.80	0.01	0.80	V
B1-B30	Input breakdown current	IV	S	0	1	0	1	0	1	0	1	0	1	mA
C1-C30	Input clamp diode voltage	V	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
D1-D8	Output voltage, low level	VI	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
E1-E30	Output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F1-F30	Input current, low level	VIII	X	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	0.40	1.73	mA
G1-G30	Input current, high level	IX	X	0	30	0	40	0	40	0	40	0	40	μA
H1-H8	Output short circuit current	X	X	20	55	20	55	20	55	20	55	20	55	mA
J1	Power supply circuit, low level	XI	S	1	44	1	44	1	44	1	44	1	44	mA
J2	Power supply circuit, high level	XI	S	1	16	1	16	1	16	1	16	1	16	mA
K1-K8	Transition time, t_{TLH}	XII	S	4	45	4	50	4	50	4	50	4	50	ns
L1-L8	Transition time, t_{THL}	XII	S	4	20	4	25	4	25	4	25	4	25	ns
M1-M8	Propagation delay time, t_{PHL}	XIII	S	4	19	4	22	4	22	4	22	4	22	ns
N1-N8	Propagation delay time, t_{PLH}	XIII	S	4	26	4	28	4	28	4	28	4	28	ns
P1-P8	Output crosstalk, low level	XIV	Q	0	450	0	450	0	450	0	450	0	450	mV
R1-R8	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
S	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ± 5°C		60°C ± 5°C		-55°C ± 5°C		85°C ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T	Power supply transient rise time	XVII	S	1	2	1	2	1	2	1	2	1	2	ms
U	Isolation													
	Initial	XVIII	S	10	-	10	-	10	-	10	-	10	-	MΩ
	End-of-life	XVIII	S	1	-	1	-	1	-	1	-	1	-	MΩ
V1-V30	Input capacitance	XIX	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, or U are part of a sequence of tests, the order of testing shall be as follows: A, B, U where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through R and T shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition T, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests C through R and T shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through V shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Cross reference of classification. Military Specification MIL-M-28787/19 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing number 10001-2833405. Part number M28787/19-1 supersedes part numbers 2833405 and M28787/19.

6.5 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

Maximum power dissipation	242 mW (all outputs low level) 88 mW (all outputs high level)
Input voltage	
Low level	0.8 V maximum
High level	2.0 V minimum

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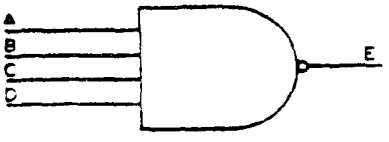
Input current	
Low level (V_{IL} = 0 V)	0.4 mA minimum, 1.73 mA maximum
(V_{IL} = 0.4 V)	1.6 mA maximum
High level (V_{IH} = 2.4 V)	0 uA minimum, 40 uA maximum
(V_{IH} = 5.5 V)	1 mA maximum
Output voltage	
Low level	0 V minimum, 0.4 V maximum
High level	2.4 V minimum, V_{CC} maximum
Output drive current	
Low level (V_{OL} = 0 4 V)	16-mA maximum
High level (V_{OH} = 2.4 V)	1 mA minimum
Propagation delay time (C_L = 100 pF)	
t_{PHL}	4 ns minimum, 22 ns maximum
t_{PLH}	4 ns minimum, 28 ns maximum

6.6 Module function. This module contains two independent 3-input and six independent 4-input NAND gates

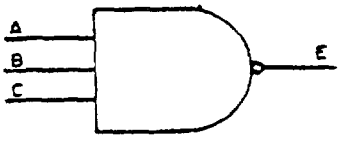
6 6.1 Input to output relationship Each NAND gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship

INPUT	OUTPUT
AT LEAST ONE INPUT = L	H
ALL INPUTS = H	L



$E = \overline{ABCD}$



$E = \overline{ABC}$

6 6 2 Parallel outputs. Paralleling of outputs is not permissible except when both inputs and outputs of circuits are paralleled to obtain extended drive. Each circuit whose output is paralleled with an output of another circuit shall have each of its inputs paralleled with a corresponding input of the other circuit. Circuits may be paralleled only with other circuits within the same group These circuit groups are

a. Circuits 1 and 2.
b. Circuits 3 and 4
c. Circuits 5 and 6.
d. Circuits 7 and 8.

The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The input loading of this combination is equal to the sum of the input loading of the individual circuits.

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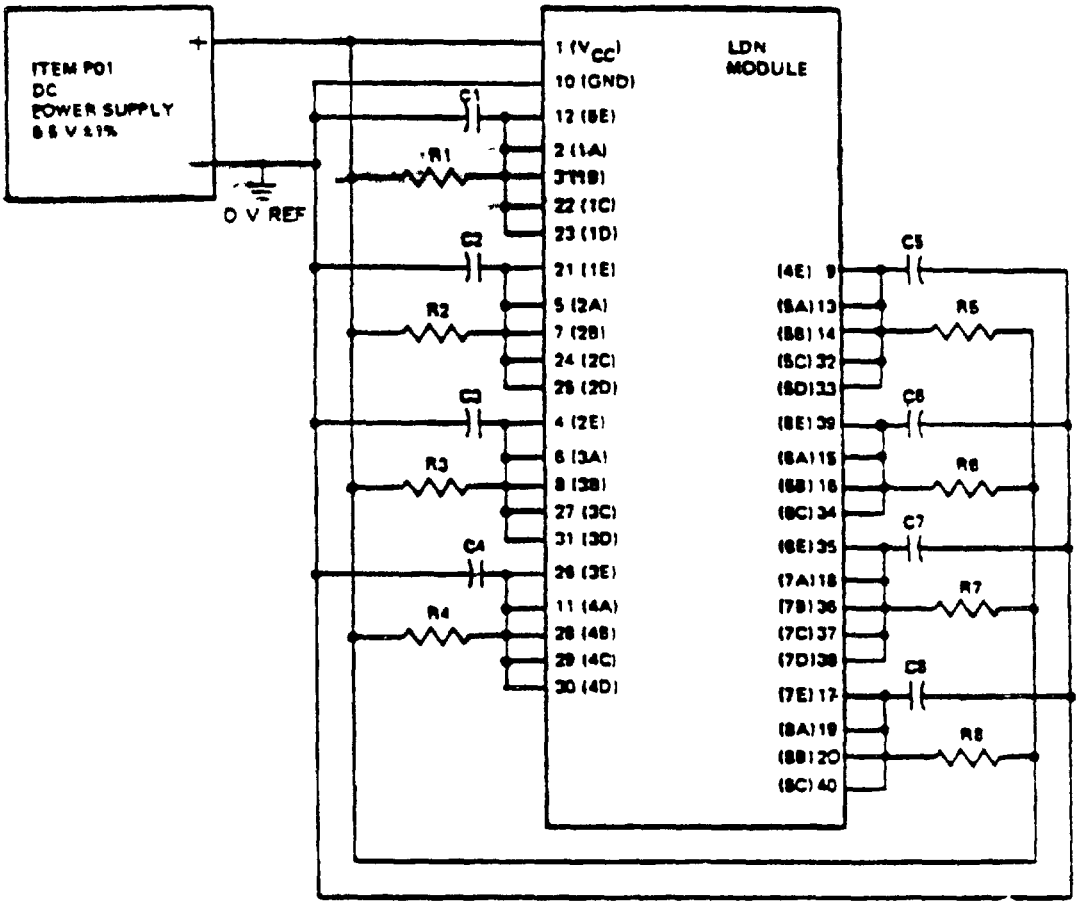
6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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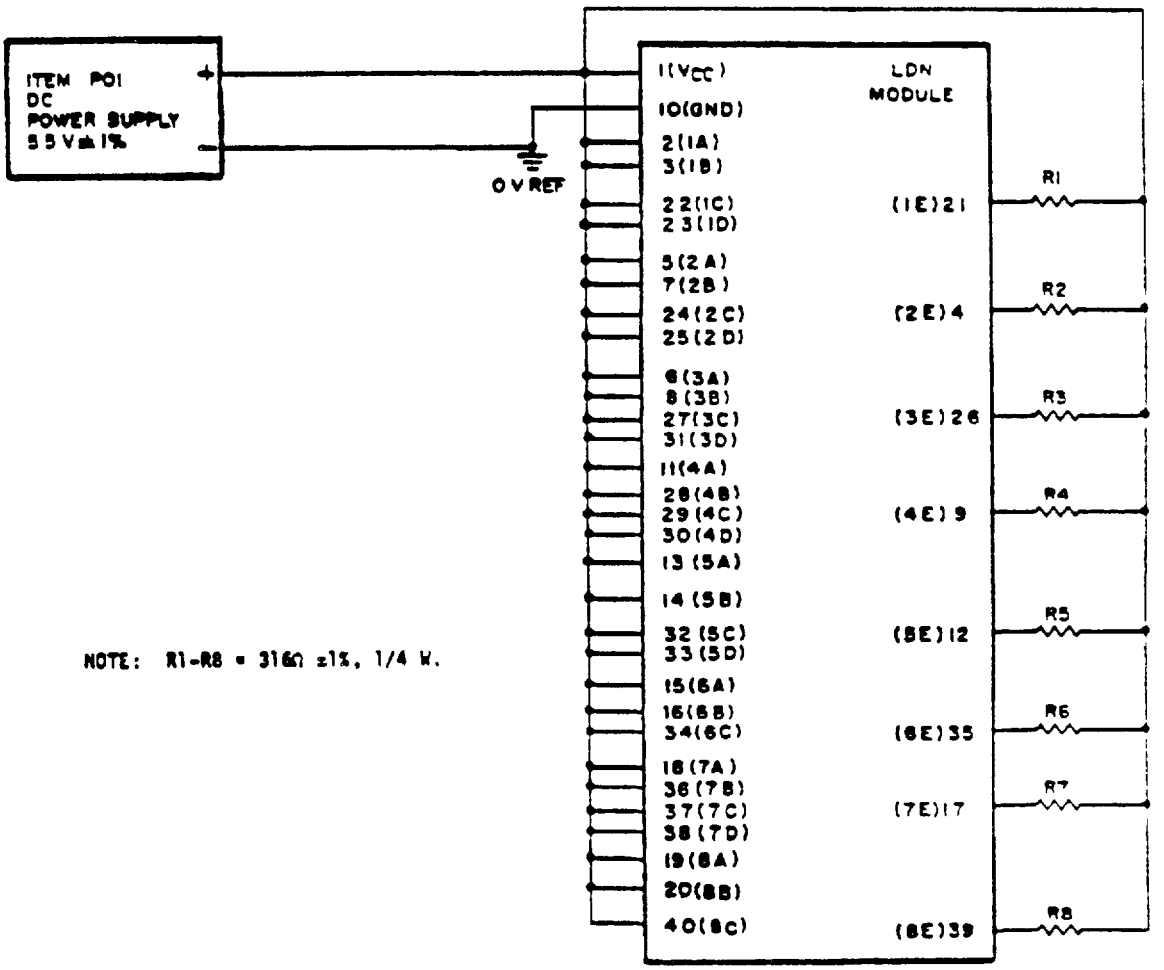
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NOTES
1 R1-R8 = 316Ω ±1%, 1/4 W.
2 C1-C8 = 100 pF ±20%.
3 When using figure 2 for module burn-in, the RC loads are not required

FIGURE 2. Operating temperature, life, and burn-in test circuit.

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NOTE: R1-R8 = 316Ω ±1%, 1/4 W.

FIGURE 3 Component test circuit

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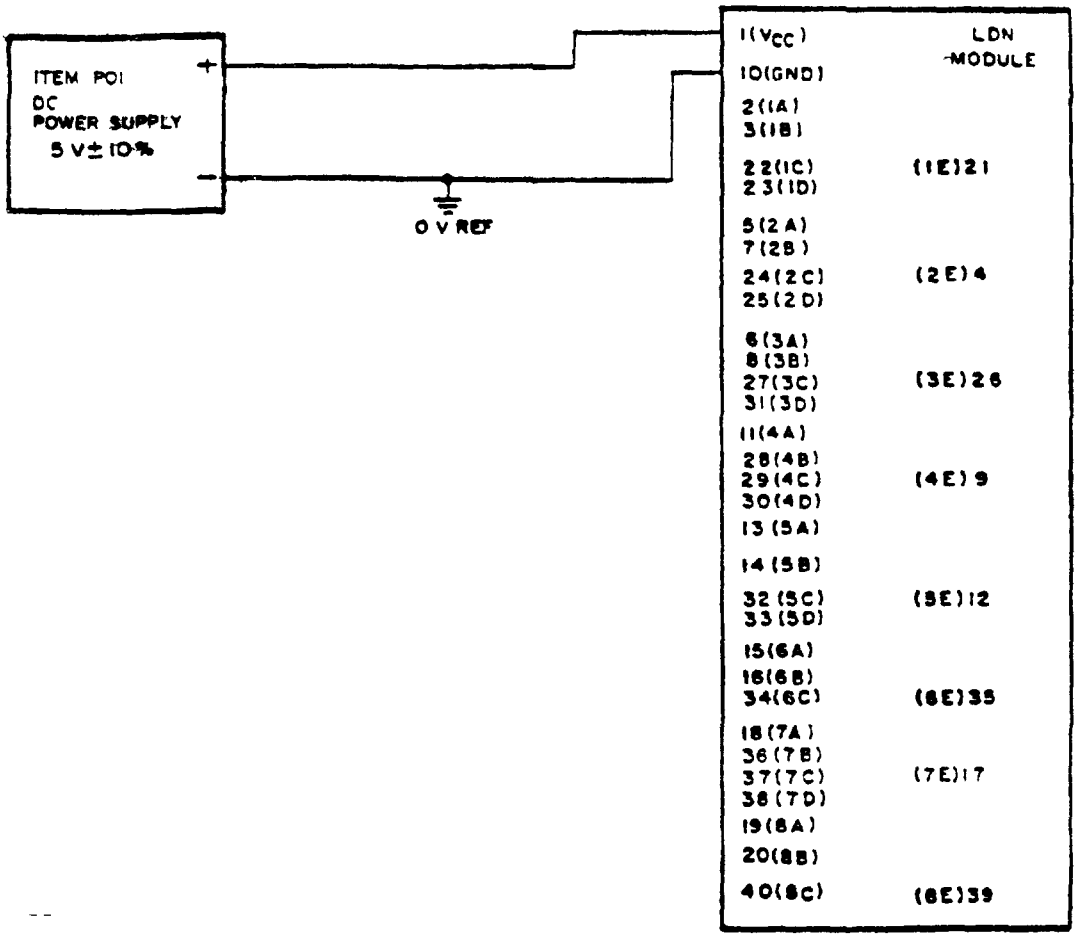


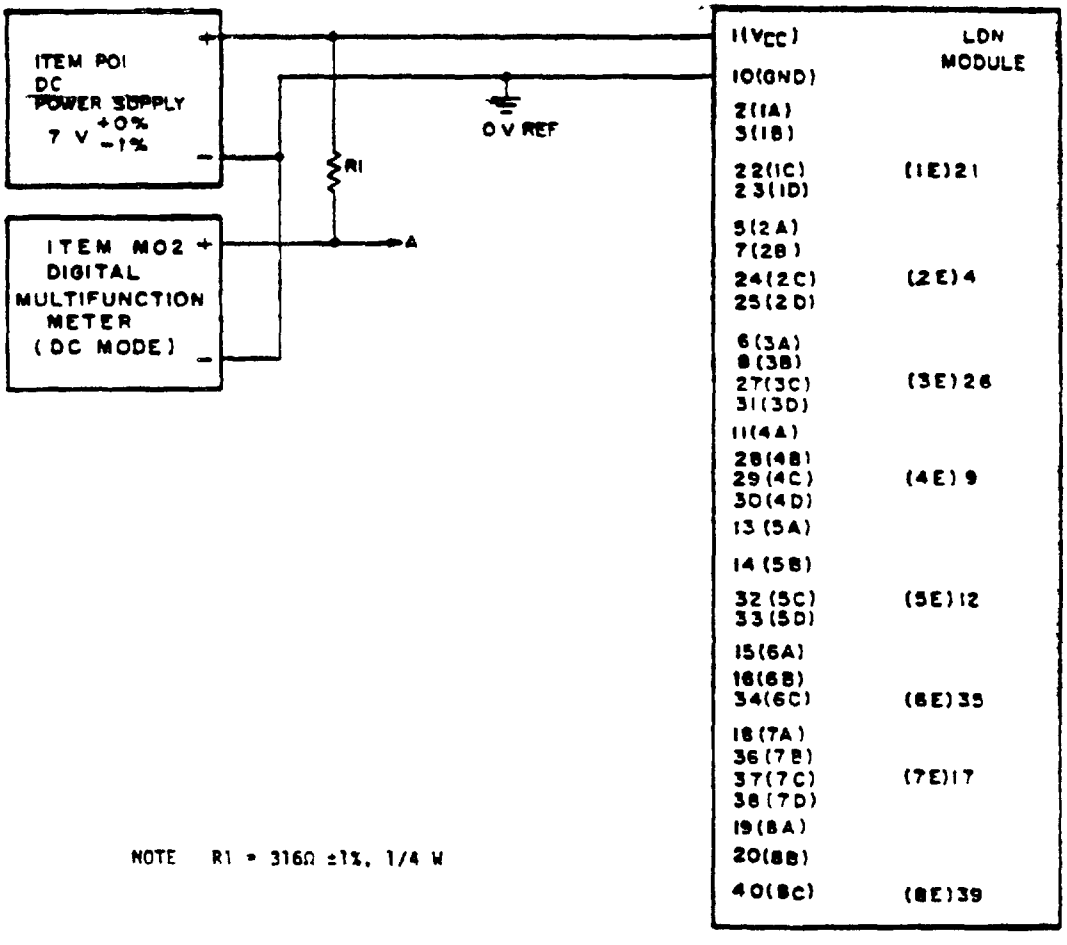
FIGURE 4. Vibration test circuit

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TABLE III. V_{OUT} (low-loaded) test.

Test condition	Parameter*	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
A1	E1 V_{OUT} (low loaded)	21	0 01	0 80	V	5
A2	E2 "	4	"	"	"	"
A3	E3 "	26	"	"	"	"
A4	E4 "	9	"	"	"	"
A5	E5 "	12	"	"	"	"
A6	E6 "	35	"	"	"	"
A7	E7 "	17	"	"	"	"
A8	E8 "	39	"	"	"	"

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NOTE R1 = 316Ω ±1%, 1/4 W

FIGURE 5. V_{OUT} (low-loaded) test circuit for table III

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TABLE IV Input breakdown current test.

Test condition	Parameter		Connect		Initial 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
B1	A1	Input breakdown current	2	3,22,23	0	1	mA	6
B2	B1	"	3	2,22,23	"	"	"	"
B3	C1	"	22	2,3,23	"	"	"	"
B4	D1	"	23	2,3,22	"	"	"	"
B5	A2	"	5	7,24,25	"	"	"	"
B6	B2	"	7	5,24,25	"	"	"	"
B7	C2	"	24	5,7,25	"	"	"	"
B8	D2	"	25	5,7,24	"	"	"	"
B9	A3	"	6	8,27,31	"	"	"	"
B10	B3	"	8	6,27,31	"	"	"	"
B11	C3	"	27	6,8,31	"	"	"	"
B12	D3	"	31	6,8,27	"	"	"	"
B13	A4	"	11	28,29,30	"	"	"	"
B14	B4	"	28	11,29,30	"	"	"	"
B15	C4	"	29	11,28,30	"	"	"	"
B16	D4	"	30	11,28,29	"	"	"	"
B17	A5	"	13	14,32,33	"	"	"	"
B18	B5	"	14	13,32,33	"	"	"	"
B19	C5	"	32	13,14,33	"	"	"	"
B20	D5	"	33	13,14,32	"	"	"	"

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TABLE IV. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B21	A6 Input breakdown current	15	16,34	0	1	mA	6
B22	B6 "	16	15,34	"	"	"	"
B23	C6 "	34	15,16	"	"	"	"
B24	A7 "	18	36,37,38	"	"	"	"
B25	B7 "	36	18,37,38	"	"	"	"
B26	C7 "	37	18,36,38	"	"	"	"
B27	D7 "	38	18,36,37	"	"	"	"
B28	A8 "	19	20,40	"	"	"	"
B29	B8 "	20	19,40	"	"	"	"
B30	C8 "	40	19,20	"	"	"	"

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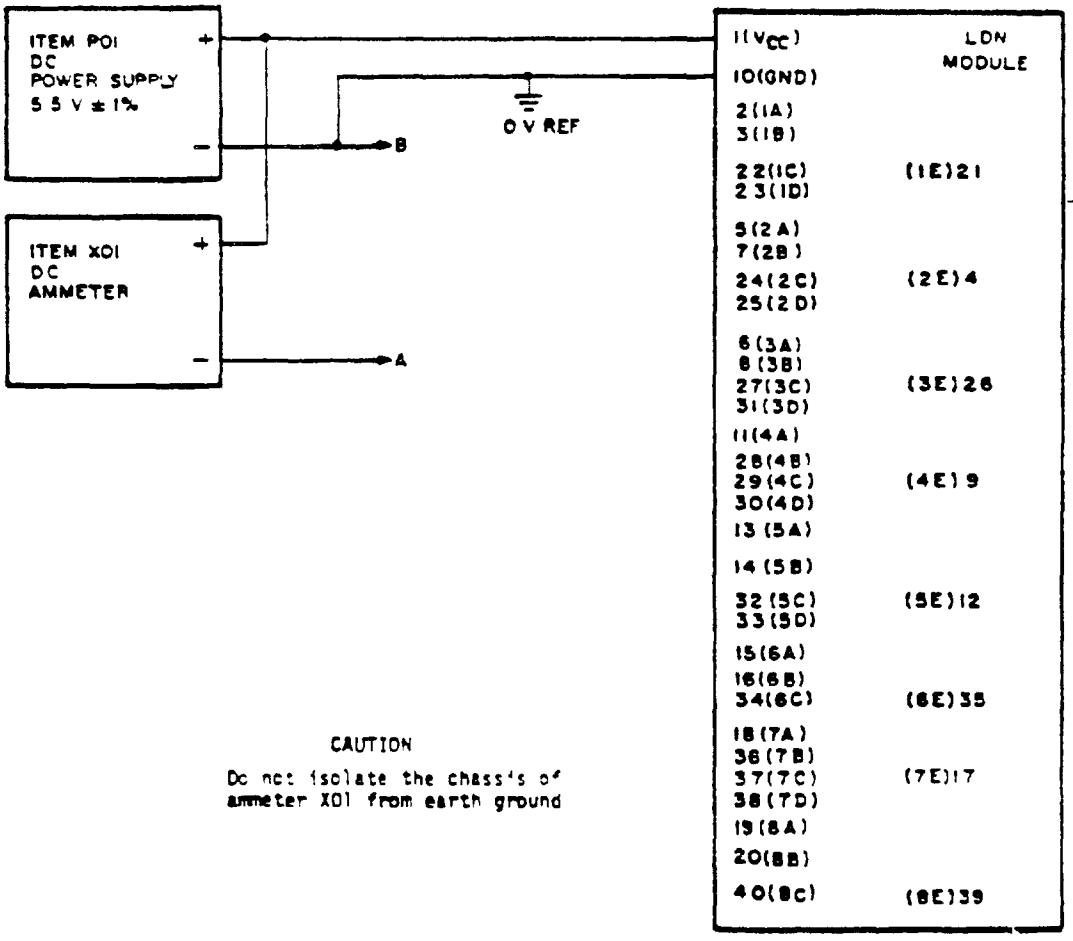


FIGURE 6 Input breakdown current test circuit for table IV

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TABLE V Input clamp diode voltage test

Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
C1	A1	Input clamp diode voltage	2	-0.4	-1.5	V	7
C2	B1	"	3	"	"	"	"
C3	C1	"	22	"	"	"	"
C4	D1	"	23	"	"	"	"
C5	A2	"	5	"	"	"	"
C6	B2	"	7	"	"	"	"
C7	C2	"	24	"	"	"	"
C8	D2	"	25	"	"	"	"
C9	A3	"	6	"	"	"	"
C10	B3	"	8	"	"	"	"
C11	C3	"	27	"	"	"	"
C12	D3	"	31	"	"	"	"
C13	A4	"	11	"	"	"	"
C14	B4	"	28	"	"	"	"
C15	C4	"	29	"	"	"	"
C16	D4	"	30	"	"	"	"
C17	A5	"	13	"	"	"	"
C18	B5	"	14	"	"	"	"
C19	C5	"	32	"	"	"	"
C20	D5	"	33	"	"	"	"

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TABLE V. Input clamp diode voltage test - Continued

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
C21	A6	Input clamp diode voltage	15	-0.4	-1.5	V	7
C22	B6	"	16	"	"	"	"
C23	C6	"	34	"	"	"	"
C24	A7	"	18	"	"	"	"
C25	B7	"	36	"	"	"	"
C26	C7	"	37	"	"	"	"
C27	D7	"	38	"	"	"	"
C28	A8	"	19	"	"	"	"
C29	B8	"	20	"	"	"	"
C30	C8	"	40	"	"	"	"

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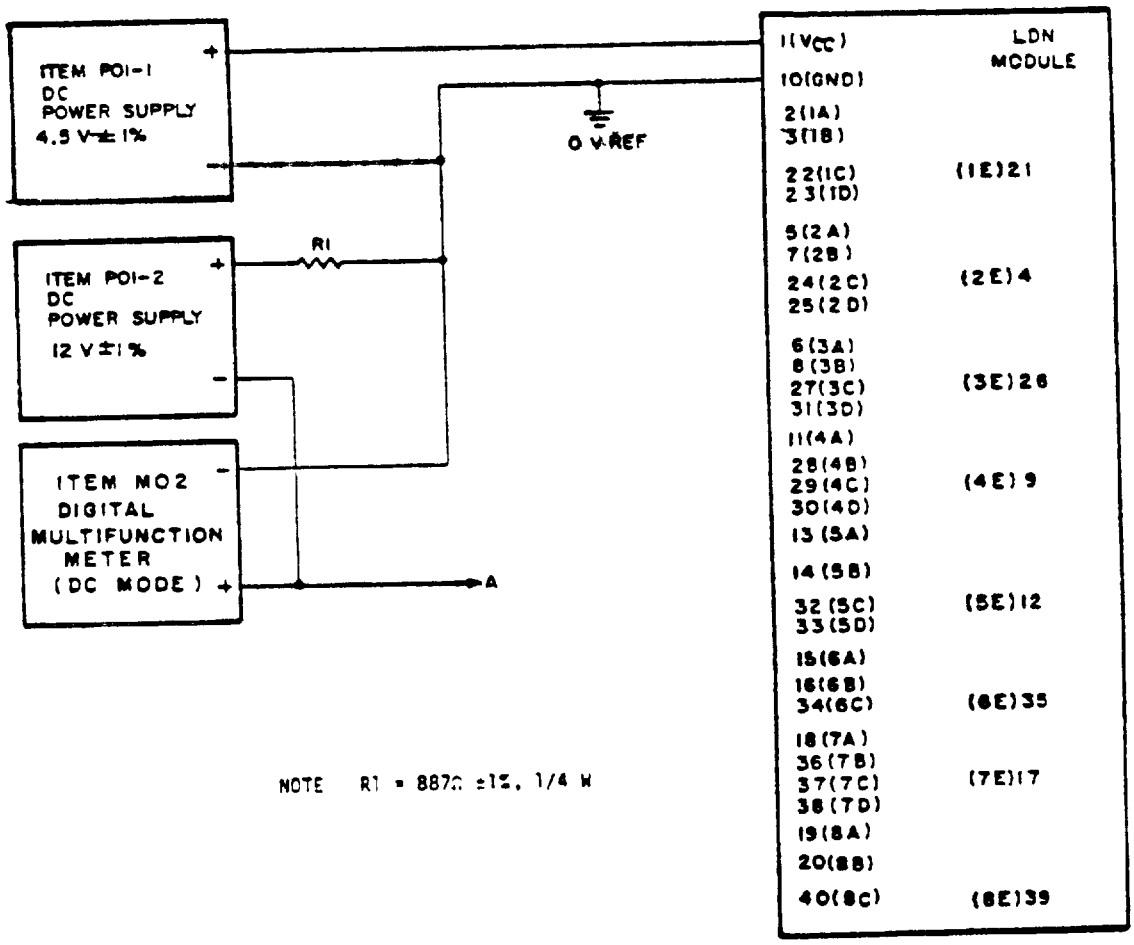


FIGURE 7. Input clamp diode voltage test circuit for table V.

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TABLE VI Output voltage (low-level) test

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
D1	E1 Output voltage, low level	2,3,22,23	Open	21	0 0	0 4	V	8
D2	E2 "	5,7,24,25	"	4	"	"	"	"
D3	E3 "	6,8,27,31	"	26	"	"	"	"
D4	E4 "	11,28,29,30	"	9	"	"	"	"
D5	E5 "	13,14,32,33	"	12	"	"	"	"
D6	E6 "	15,16,34	"	35	"	"	"	"
D7	E7 "	18,36,37,38	"	17	"	"	"	"
D8	E8 "	19,20,40	"	39	"	"	"	"

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TABLE VII. Output voltage (high-level) test.

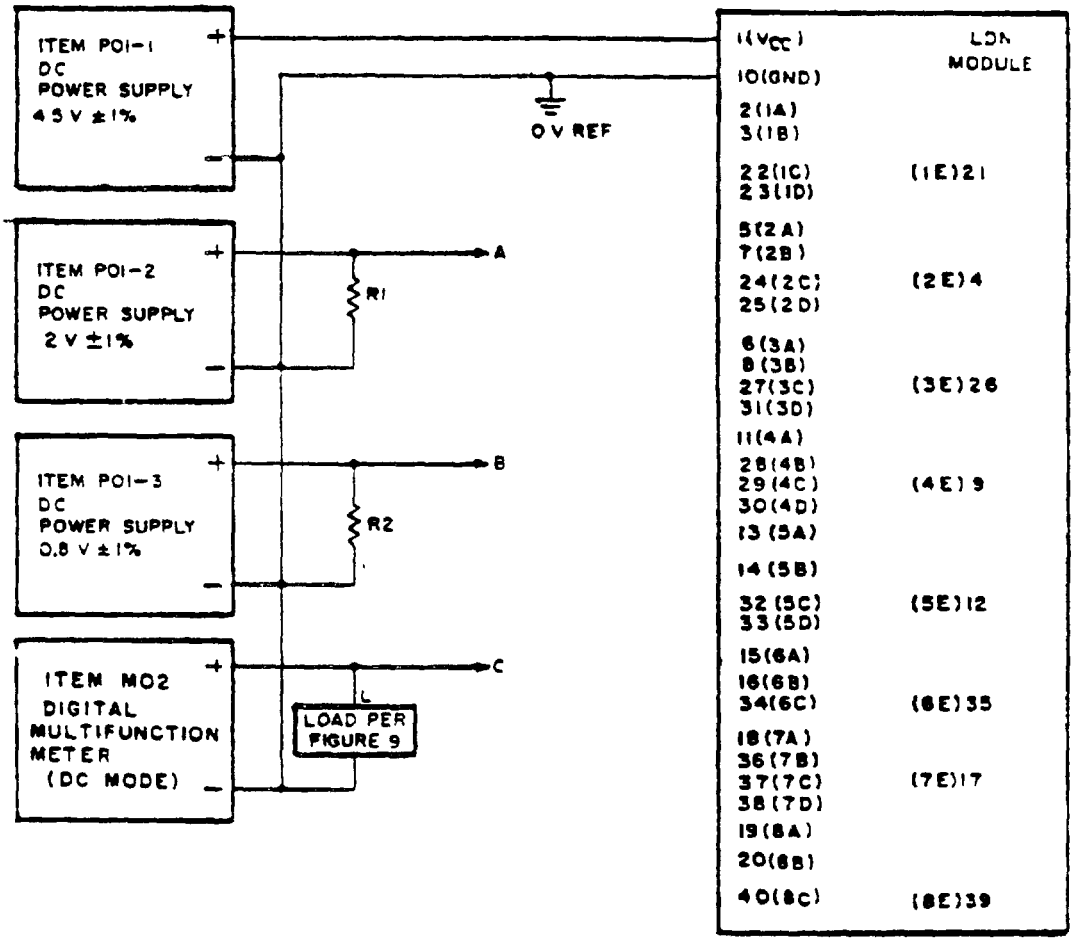
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E1	E1 Output voltage, high level	Open	2	21	2.4	4.5	V	8
E2	E1 "	"	3	"	"	"	"	"
E3	E1 "	"	22	"	"	"	"	"
E4	E1 "	"	23	"	"	"	"	"
E5	E2 "	"	5	4	"	"	"	"
E6	E2 "	"	7	"	"	"	"	"
E7	E2 "	"	24	"	"	"	"	"
E8	E2 "	"	25	"	"	"	"	"
E9	E3 "	"	6	26	"	"	"	"
E10	E3 "	"	8	"	"	"	"	"
E11	E3 "	"	27	"	"	"	"	"
E12	E3 "	"	31	"	"	"	"	"
E13	E4 "	"	11	9	"	"	"	"
E14	E4 "	"	28	"	"	"	"	"
E15	E4 "	"	29	"	"	"	"	"
E16	E4 "	"	30	"	"	"	"	"
E17	E5 "	"	13	12	"	"	"	"
E18	E5 "	"	14	"	"	"	"	"
E19	E5 "	"	32	"	"	"	"	"
E20	E5 "	"	33	"	"	"	"	"

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TABLE VII Output voltage (high-level) test - Continued

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
E21	E6 Output voltage, high level	Open	15	35	2.4	4.5	V	8
E22	E6 "	"	16	"	"	"	"	"
E23	E6 "	"	34	"	"	"	"	"
E24	E7 "	"	18	17	"	"	"	"
E25	E7 "	"	36	"	"	"	"	"
E26	E7 "	"	37	"	"	"	"	"
E27	E7 "	"	38	"	"	"	"	"
E28	E8 "	"	19	39	"	"	"	"
E29	E8 "	"	20	"	"	"	"	"
E30	E8 "	"	40	"	"	"	"	"

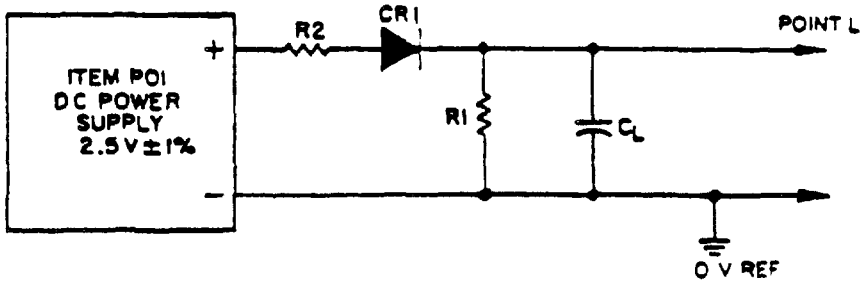
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NOTE: R1 = R2 = 51Ω ±5%, 1 W

FIGURE 8 Output voltage (low-level and high-level) test circuit for tables VI and VII.

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NOTES.
1. R1 = Select value for a current flow of 1 mA ±1% out of point L with point L held at 2.4 V dc.
2. R2 = Select value for a current flow of 16 mA ±1% with point L held at 0.4 V.
3. CR1 = 1N4150 or equivalent.
4. C_L = 100 ±5 pF (including probe and parasitic capacitance) for test conditions K, L, M, and N.
5. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions P and R.
6. C_L is not required for dc electrical tests D1-D8 or E1-E30.
7. The maximum lead length from point L to the appropriate module pin shall be 3 inches (76.2 mm).
8 During output switching, the 2.5 V power supply shall have a maximum overshoot of 500 mV.

FIGURE 9. Logic load circuit.

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TABLE VIII. Input current (low-level) test

Test condition	Parameter	Connect			Initial limits 25°C 45°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F1	A1 Input current, low level	3,22,23	2	Open	0 40	1 73	mA	10
F2	B1 "	2,22,23	3	"	"	"	"	"
F3	C1 "	2,3,23	22	"	"	"	"	"
F4	D1 "	2,3,22	23	"	"	"	"	"
F5	A2 "	7,24,25	5	"	"	"	"	"
F6	B2 "	5,24,25	7	"	"	"	"	"
F7	C2 "	5,7,25	24	"	"	"	"	"
F8	D2 "	5,7,24	25	"	"	"	"	"
F9	A3 "	8,27,31	6	"	"	"	"	"
F10	B3 "	6,27,31	8	"	"	"	"	"
F11	C3 "	6,8,31	27	"	"	"	"	"
F12	D3 "	6,8,27	31	"	"	"	"	"
F13	A4 "	28,29,10	11	"	"	"	"	"
F14	B4 "	11,29,30	28	"	"	"	"	"
F15	C4 "	11,28,30	29	"	"	"	"	"
F16	D4 "	11,28,29	30	"	"	"	"	"
F17	A5 "	14,32,33	13	"	"	"	"	"
F18	B5 "	13,32,33	14	"	"	"	"	"
F19	C5 "	13,14,33	32	"	"	"	"	"
F20	D5 "	13,14,32	33	"	"	"	"	"

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TABLE VIII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
F21	A6 Input current, low level	16,34	15	Open	0.40	1.73	mA	10
F22	B6 "	15,34	16	"	"	"	"	"
F23	C6 "	15,16	34	"	"	"	"	"
F24	A7 "	36,37,38	18	"	"	"	"	"
F25	B7 "	18,37,38	36	"	"	"	"	"
F26	C7 "	18,36,38	37	"	"	"	"	"
F27	D7 "	18,36,37	38	"	"	"	"	"
F28	A8 "	20,40	19	"	"	"	"	"
F29	B8 "	19,40	20	"	"	"	"	"
F30	C8 "	19,20	40	"	"	"	"	"

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TABLE IX. Input current (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	A1 Input current, high level	2	Open	3,22,23	0	30	µA	10
G2	B1 "	3	"	2,22,23	"	"	"	"
G3	C1 "	22	"	2,3,23	"	"	"	"
G4	D1 "	23	"	2,3,22	"	"	"	"
G5	A2 "	5	"	7,24,25	"	"	"	"
G6	B2 "	7	"	5,24,25	"	"	"	"
G7	C2 "	24	"	5,7,25	"	"	"	"
G8	D2 "	25	"	5,7,24	"	"	"	"
G9	A3 "	6	"	8,27,31	"	"	"	"
G10	B3 "	8	"	6,27,31	"	"	"	"
G11	C3 "	27	"	6,8,31	"	"	"	"
G12	D3 "	31	"	6,8,27	"	"	"	"
G13	A4 "	11	"	28,29,30	"	"	"	"
G14	B4 "	28	"	11,29,30	"	"	"	"
G15	C4 "	29	"	11,28,30	"	"	"	"
G16	D4 "	30	"	11,28,29	"	"	"	"
G17	A5 "	13	"	14,32,33	"	"	"	"
G18	B5 "	14	"	13,32,33	"	"	"	"
G19	C5 "	32	"	13,14,33	"	"	"	"
G20	D5 "	33	"	13,14,32	"	"	"	"

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TABLE IX Input current (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G21	A6 Input current, high level	15	Open	16,34	0	30	µA	10
G22	B6 "	16	"	15,34	"	"	"	"
G23	C6 "	34	"	15,16	"	"	"	"
G24	A7 "	18	"	36,37,38	"	"	"	"
G25	B7 "	36	"	18,37,38	"	"	"	"
G26	C7 "	37	"	18,36,38	"	"	"	"
G27	D7 "	38	"	18,36,37	"	"	"	"
G28	A8 "	19	"	20,40	"	"	"	"
G29	B8 "	20	"	19,40	"	"	"	"
G30	C8 "	40	"	19,20	"	"	"	"

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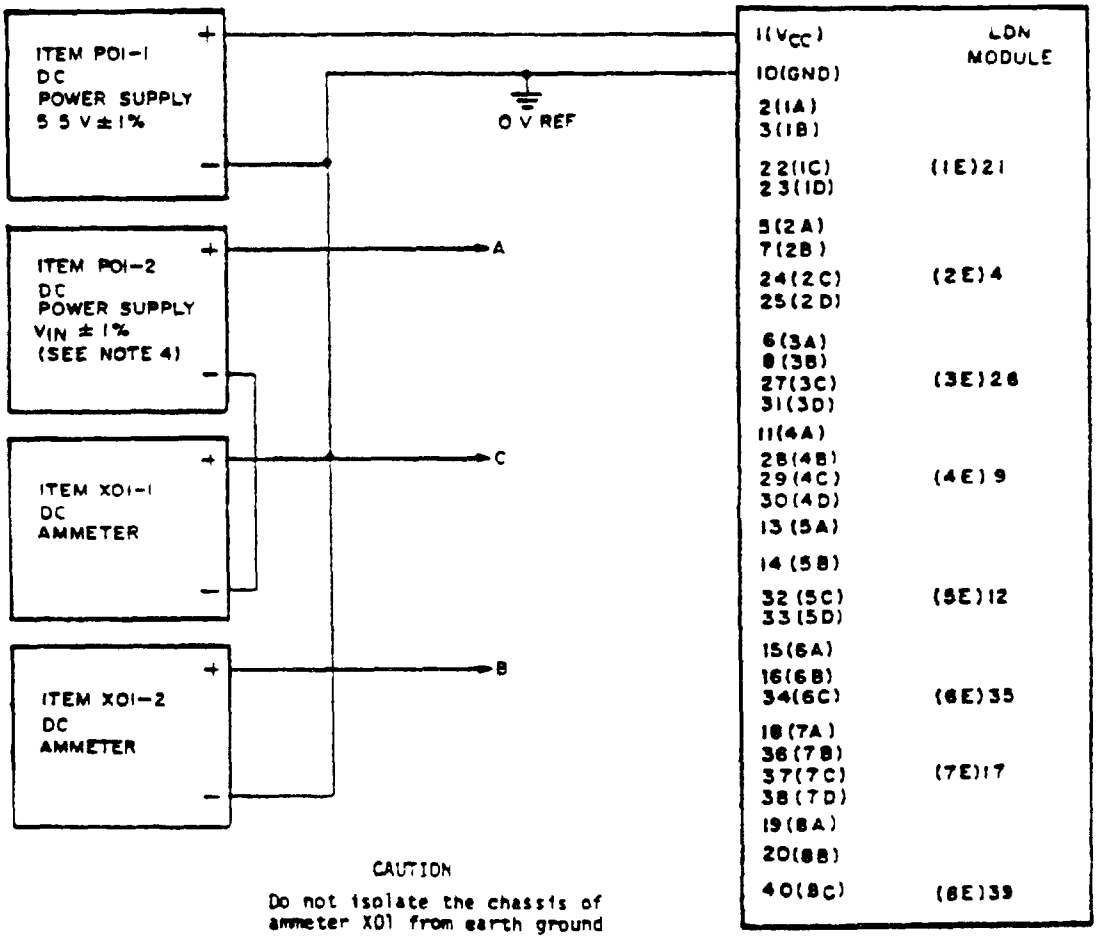
TABLE X. Output short circuit current test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	E1 Output short circuit current	Open	21	2,3,22,23	20	55	mA	10
H2	E2 "	"	4	5,7,24,25	"	"	"	"
H3	E3 "	"	26	6,8,27,31	"	"	"	"
H4	E4 "	"	9	11,28,29,30	"	"	"	"
H5	E5 "	"	12	13,14,32,33	"	"	"	"
H6	E6 "	"	35	15,16,34	"	"	"	"
H7	E7 "	"	17	18,36,37,38	"	"	"	"
H8	E8 "	"	39	19,20,40	"	"	"	"

TABLE XI. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	Power supply current, low level	Open	Open	1	44	mA	11
J2	Power supply current, high level	8	A	1	16	mA	11

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NOTES
1. Since items X01-1 and X01-2 are not used simultaneously, only one item X01 is required. Use item X01-2 for tables VIII and X, use item X01-1 for table IX.
2 With point A disconnected, item X01-1 should indicate zero current on the 1 uA range
3. When performing output short circuit tests, only one output shall be shorted at a time and the measurement taken within 5 seconds.
4. For test conditions F1-F30, V_{IN} = 4.5 V. For test conditions G1-G30, V_{IN} = 2.4 V.

FIGURE 10. Input current (low-level and high-level) and output short circuit current test circuit for tables VIII, IX, and X.

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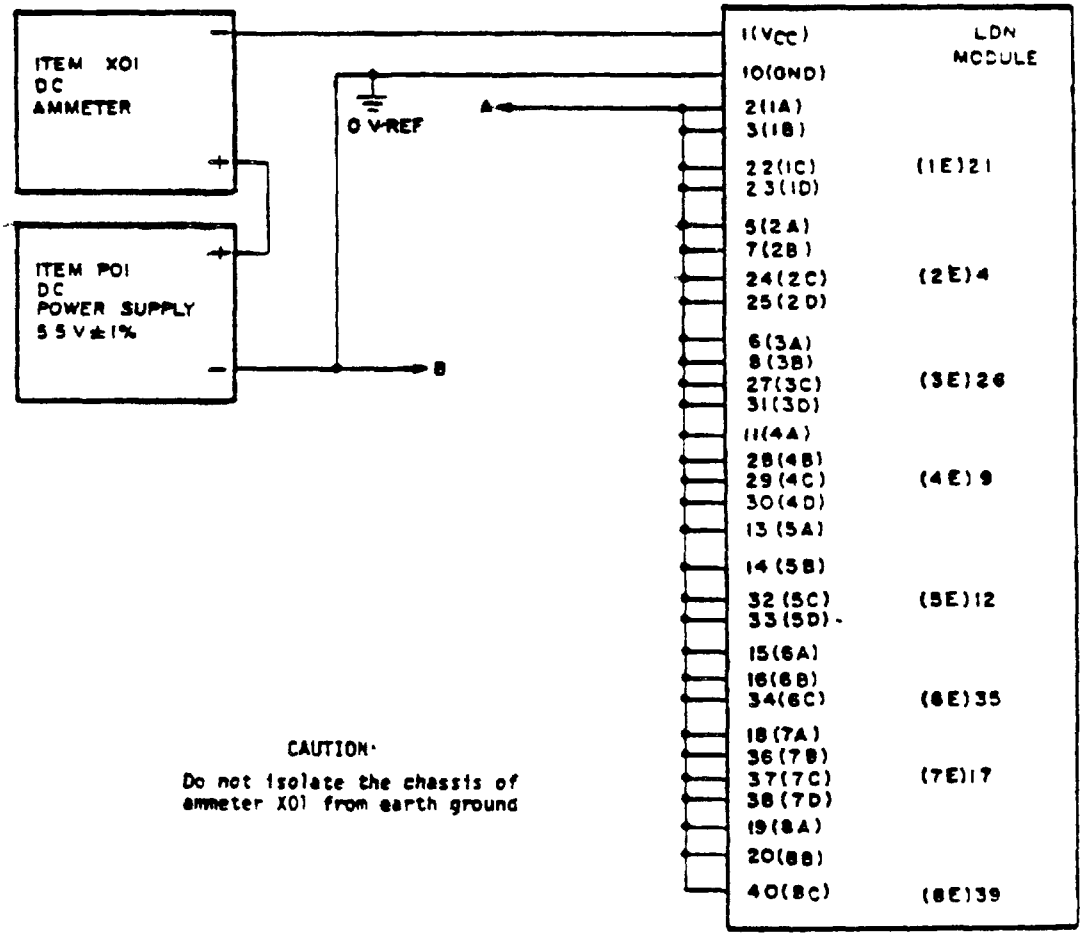


FIGURE 11. Power supply current (low-level and high-level) test circuit for table XI

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TABLE XII. Transition time (t_{TLH} and t_{THL}) test.

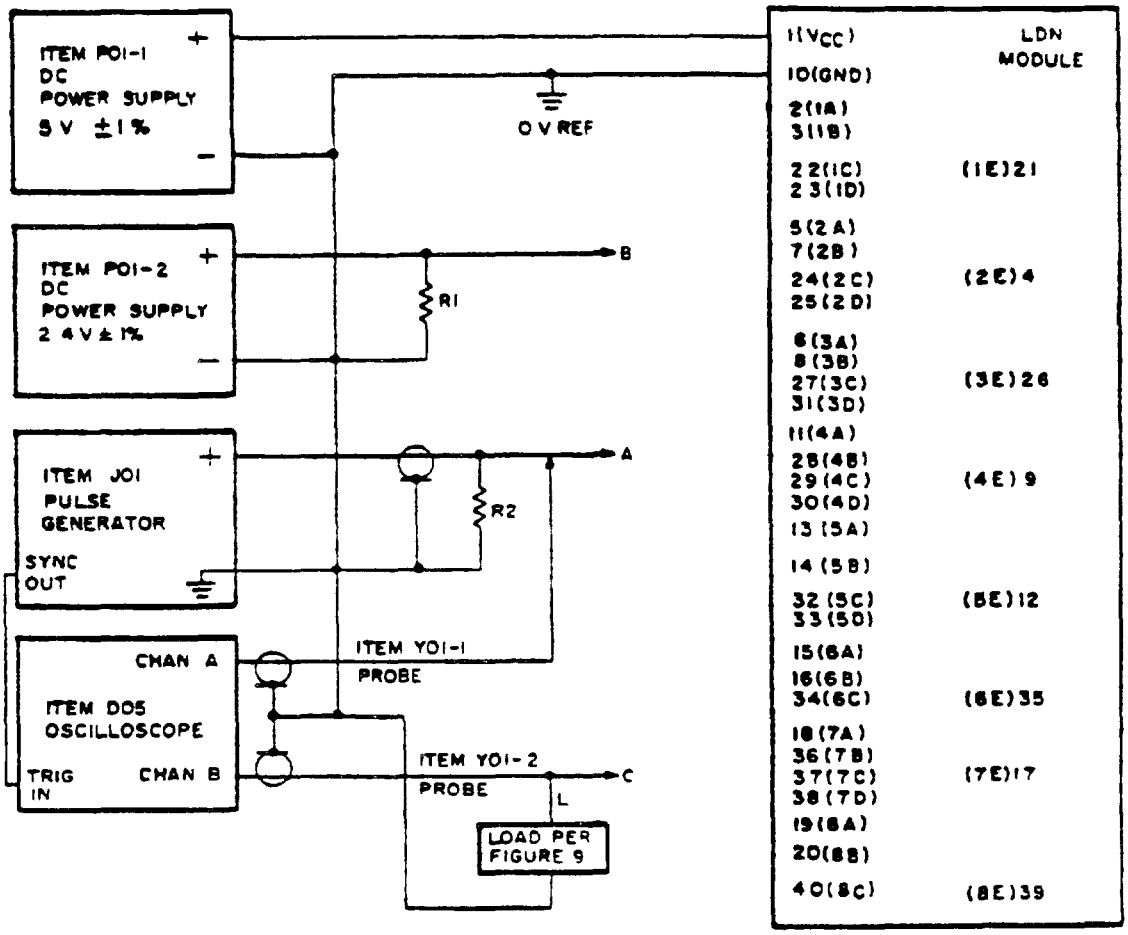
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient				Unit	Test figure
		A to pin	B to pin	C to pin	t_{TLH}		t_{THL}			
					Min	Max	Min	Max		
K1,L1	E1 Transition time	2	3,22,23	21	4	45	4	20	ns	12,13
K2,L2	E2 "	5	7,24,25	4	"	"	"	"	"	"
K3,L3	E3 "	6	8,27,31	26	"	"	"	"	"	"
K4,L4	E4 "	11	28,29,30	9	"	"	"	"	"	"
K5,L5	E5 "	13	14,32,33	12	"	"	"	"	"	"
K6,L6	E6 "	15	16,34	35	"	"	"	"	"	"
K7,L7	E7 "	18	36,37,38	17	"	"	"	"	"	"
K8,L8	E8 "	19	20,40	39	"	"	"	"	"	"

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TABLE XIII. Propagation delay time (t_{PHL} and t_{PLH}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient				Unit	Test figure
		A to pin	B to pin	C to pin	t_{PHL}		t_{PLH}			
					Min	Max	Min	Max		
M1,N1	E1 Propagation delay time	2	3,22,23	21	4	19	4	26	ns	12,13
M2,N2	E2 "	5	7,24,25	4	"	"	"	"	"	"
M3 N3	E3 "	6	8,27,31	26	"	"	"	"	"	"
M4 N4	E4 "	11	28,29,30	9	"	"	"	"	"	"
M5,N5	E5 "	13	14,32,33	12	"	"	"	"	"	"
M6,N6	E6 "	15	16,34	35	"	"	"	"	"	"
M7,N7	E7 "	18	36,37,38	17	"	"	"	"	"	"
M8,N8	E8 "	19	20,40	39	"	"	"	"	"	"

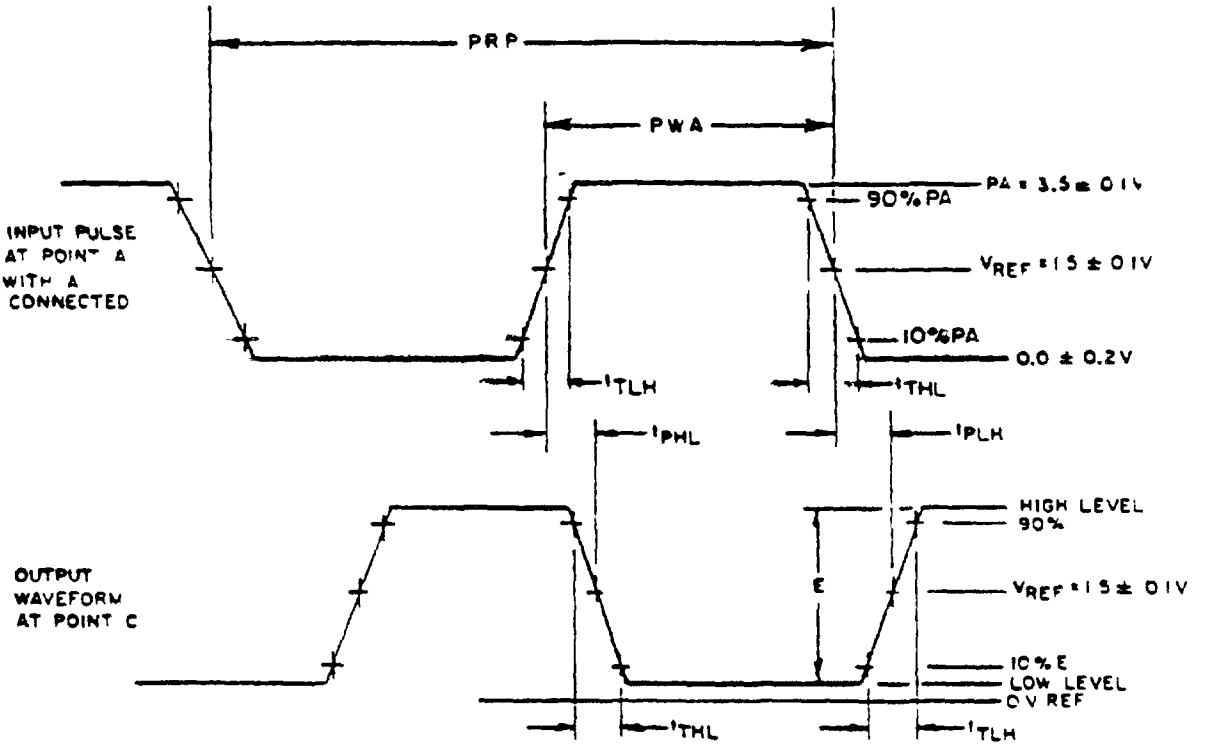
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NOTE R1 = R2 = 51Ω ±5%, 1 W.

FIGURE 12 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit for tables XII and XIII

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NOTES
1 Input pulse parameters
PWA = 100 ns ±10%
$t_{TLH} = t_{THL} = 10 \pm 2$ns
PRP = 200 ns ±10%.
2 Verify that all outputs have a high level $\geq$ 2.4 V and a low level $\leq$ 0.4 V before making any measurements
3 All voltage levels are referenced to ground

FIGURE 13 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PHL} and t_{PLH}) test circuit waveforms for tables XII and XIII.

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TABLE XIV Output crosstalk (low-level) test

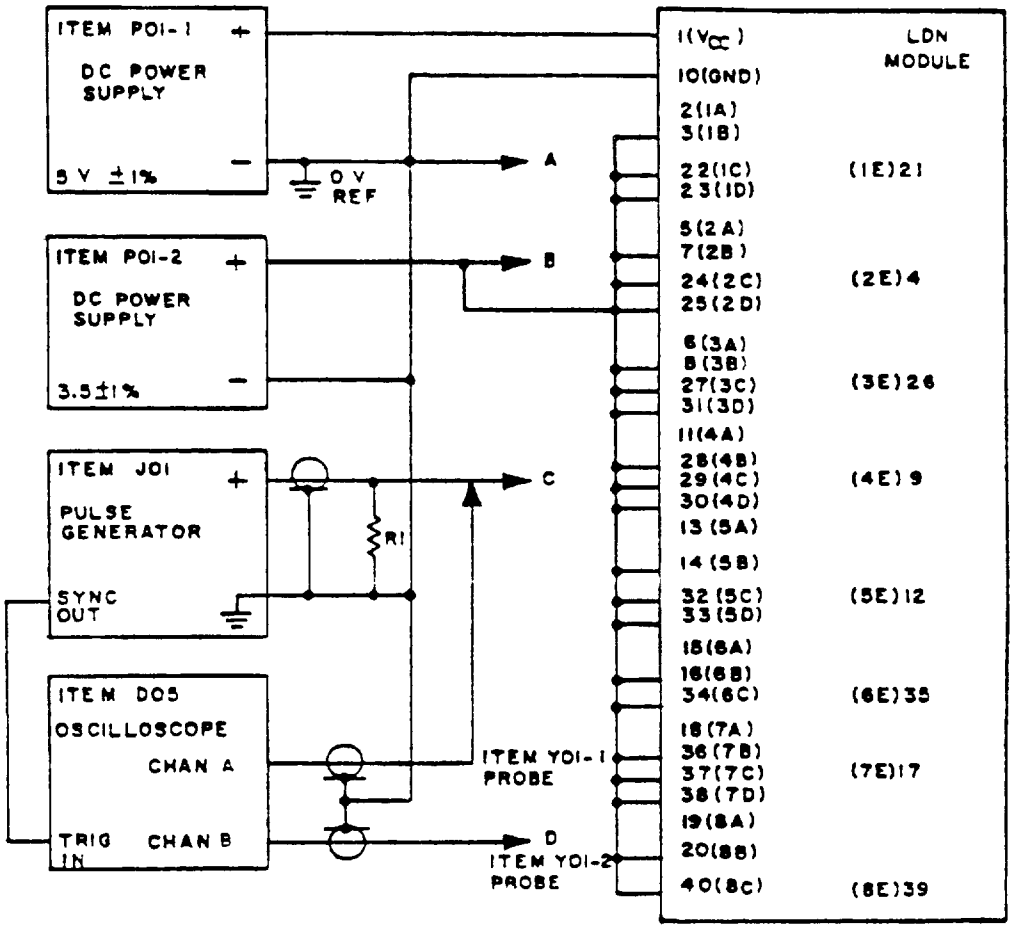
Test condition	Parameter	Connect				Initial limits 25°C 45°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P1	E1 Output crosstalk, low level	Open	2	5,6,11,13,15,18,19	21	0	450	mV	14,15
P2	E2 "	"	5	2,6,11,13,15,18,19	4	"	"	"	"
P3	E3 "	"	6	2,5,11,13,15,18,19	26	"	"	"	"
P4	E4 "	"	11	2,5,6,13,15,18,19	9	"	"	"	"
P5	E5 "	"	13	2,5,6,11,15,18,19	12	"	"	"	"
P6	E6 "	"	15	2,5,6,11,13,18,19	35	"	"	"	"
P7	E7 "	"	18	2,5,6,11,13,15,19	17	"	"	"	"
P8	E8 "	"	19	2,5,6,11,13,15,18	39	"	"	"	"

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TABLE XV Output crosstalk (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C 45°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	E1 Output crosstalk, high level	2	Open	5,6,11,13,15,18,19	21	0	400	mV	14,15
R2	E2 "	5	"	2,6,11,13,15,18,19	4	"	"	"	"
R3	E3 "	6	"	2,5,11,13,15,18,19	26	"	"	"	"
R4	E4 "	11	"	2,5,6,13,15,18,19	9	"	"	"	"
R5	E5 "	13	"	2,5,6,11,15,18,19	12	"	"	"	"
R6	E6 "	15	"	2,5,6,11,13,18,19	35	"	"	"	"
R7	E7 "	18	"	2,5,6,11,13,15,19	17	"	"	"	"
R8	E8 "	19	"	2,5,6,11,13,15,18	39	"	"	"	"

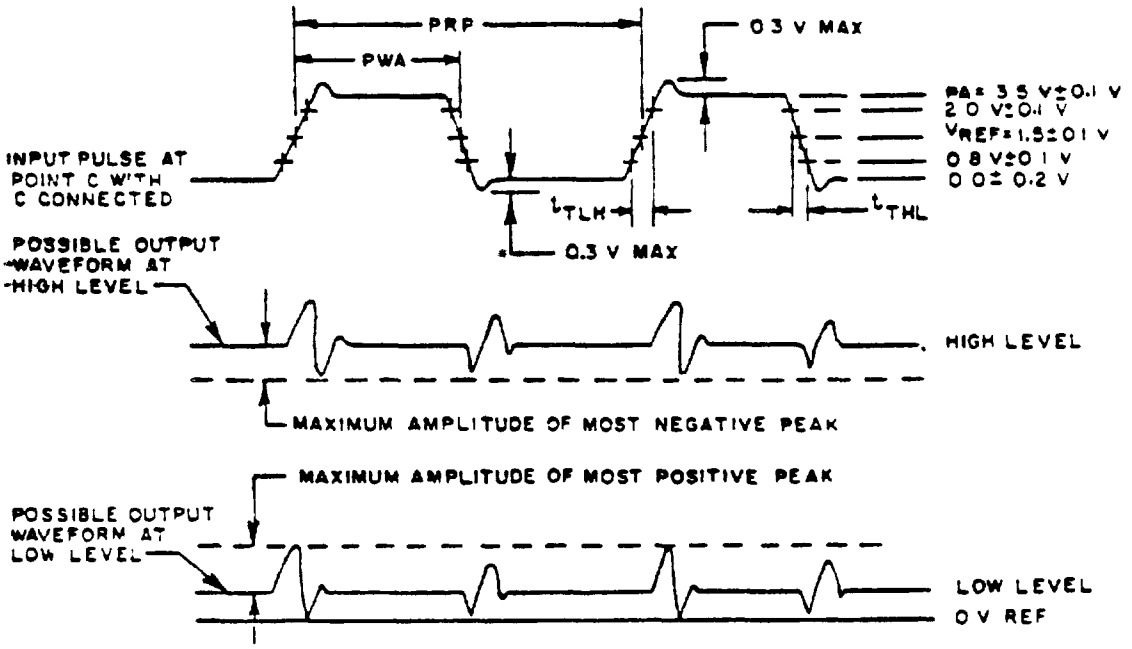
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NOTES
1 R1 = 51Ω ±5%, 1 W
2 Load each output with load as specified on figure 9.

FIGURE 14 Output crosstalk (low-level and high-level) test circuit for tables XIV and XV

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NOTES
1. Input pulse parameters:
   $t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns (verified at all input pins).
   PWA = 200 ±20 ns.
   PRP = 1000 ±100 ns.
2. The lead length of electrical conductors between resistor R1 and input pins, probes Y01-1 and Y01-2 to input and output pins, load circuits to output pins, and ground leads of probes Y01-1 and Y01-2 to test fixture and module ground pins, shall be less than 2 inches (50.8 mm)

FIGURE 15 Output crosstalk (low-level and high-level) test circuit waveforms for tables XIV and XV

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TABLE XVI Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
S	Power supply transient amplitude	0	400	mV	16,17

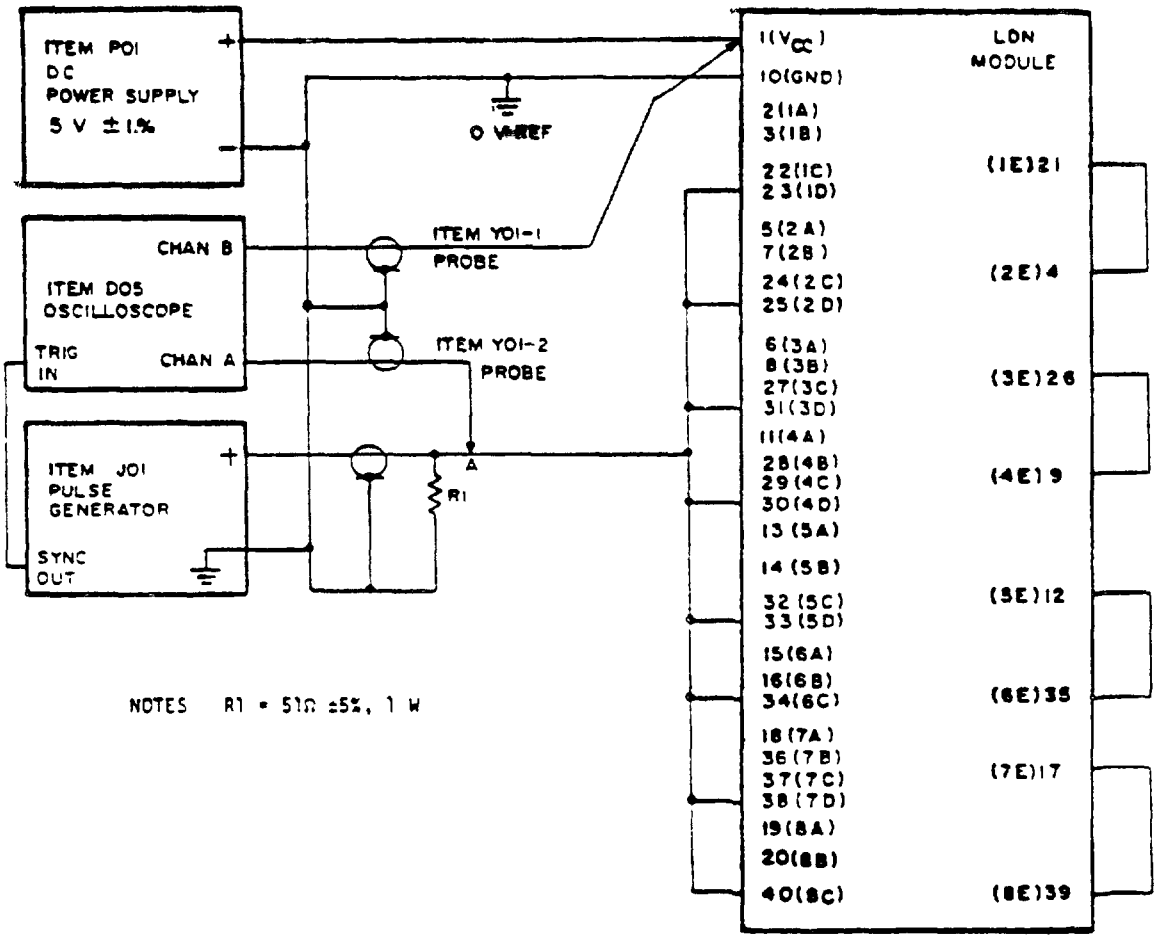
TABLE XVII. Power supply transient rise time.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient rise time	1	2	ms	18,19

TABLE XVIII. Isolation test.

Test condition	Parameter	Connect megohm bridge item R06		Initial limits 25°C ±5°C ambient		Unit	Test figure
		Positive terminal	Negative terminal	Min	Max		
U	Isolation	All pins	Frame	10	-	MΩ	20

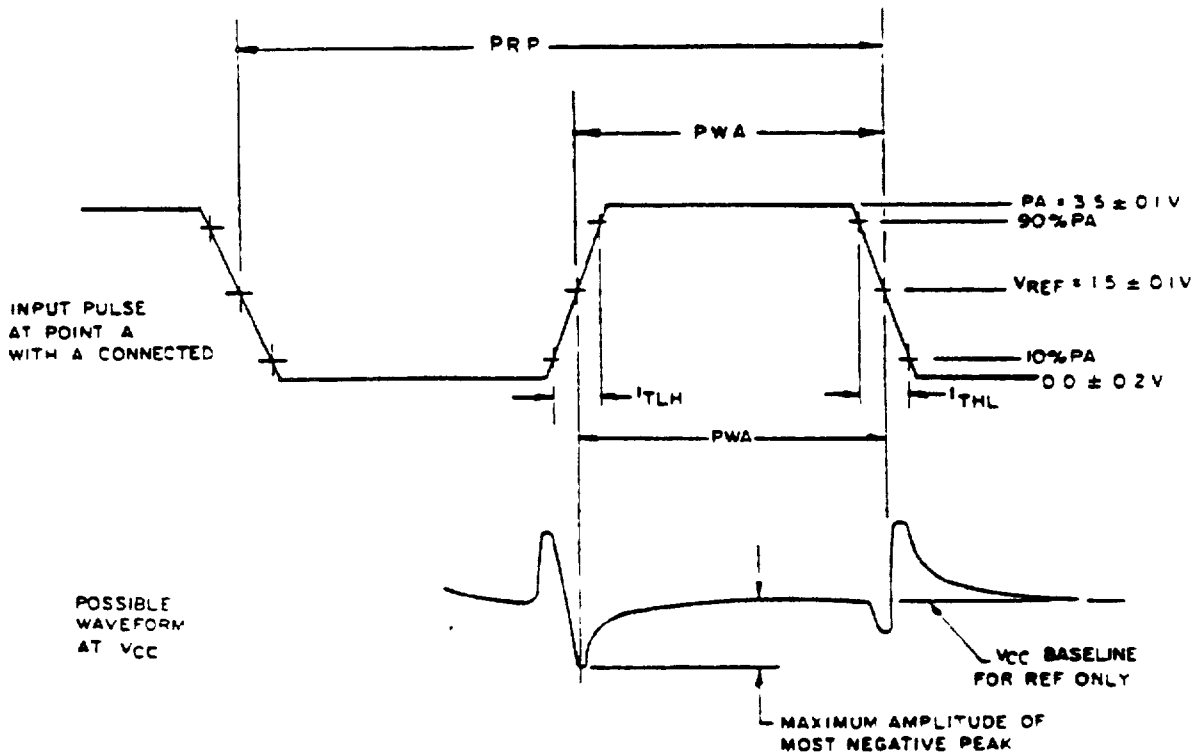
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NOTES R1 = 51Ω ±5%, 1 W

FIGURE 16 Power supply transient amplitude test circuit for table XVI.

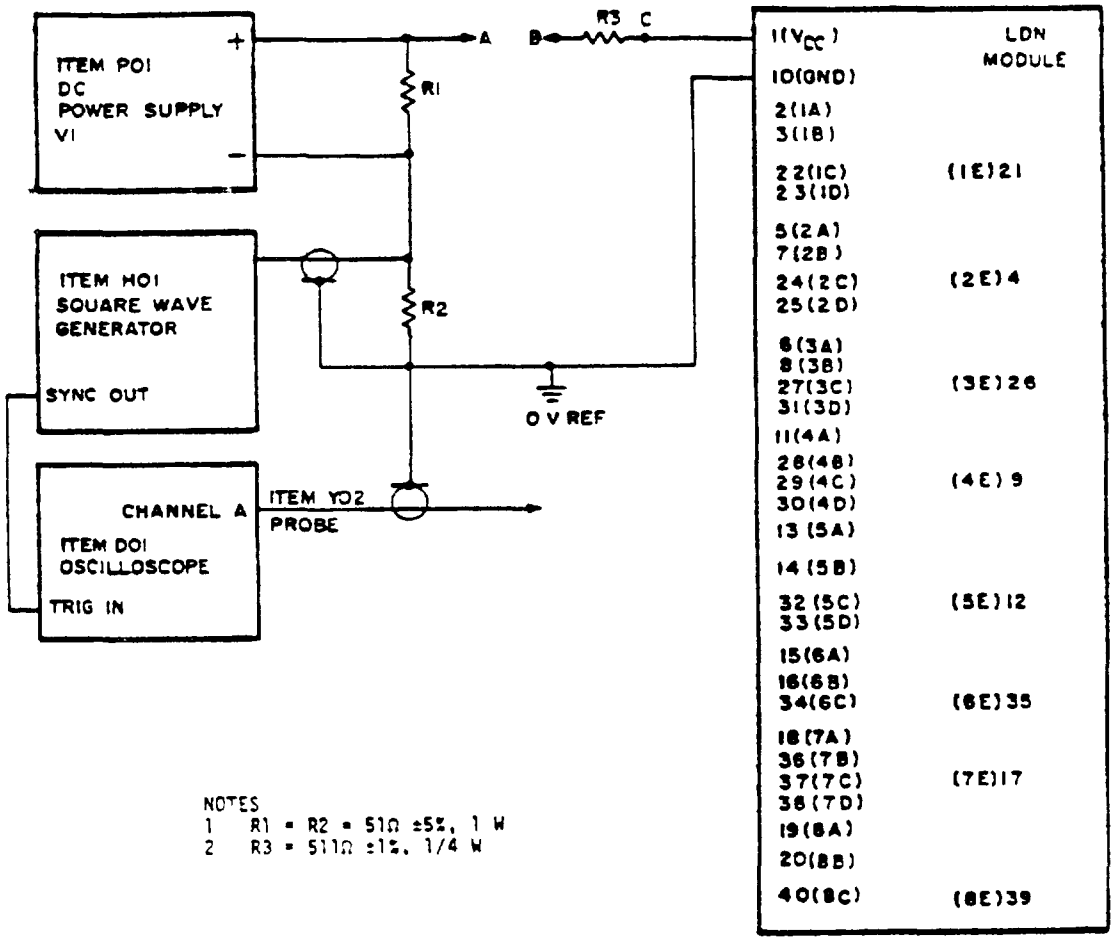
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NOTES
1. Input pulse parameters
   PWA = 3 µs ±10%.
   $t_{TLH} = t_{THL} = 10 \pm 1$ ns.
   PRP = 10 µs ±10%.
2. All voltage levels are referenced to ground
3. Measurement shall be made from the baseline to the most negative peak.

FIGURE 17 Power supply transient amplitude test circuit waveforms for table XVI

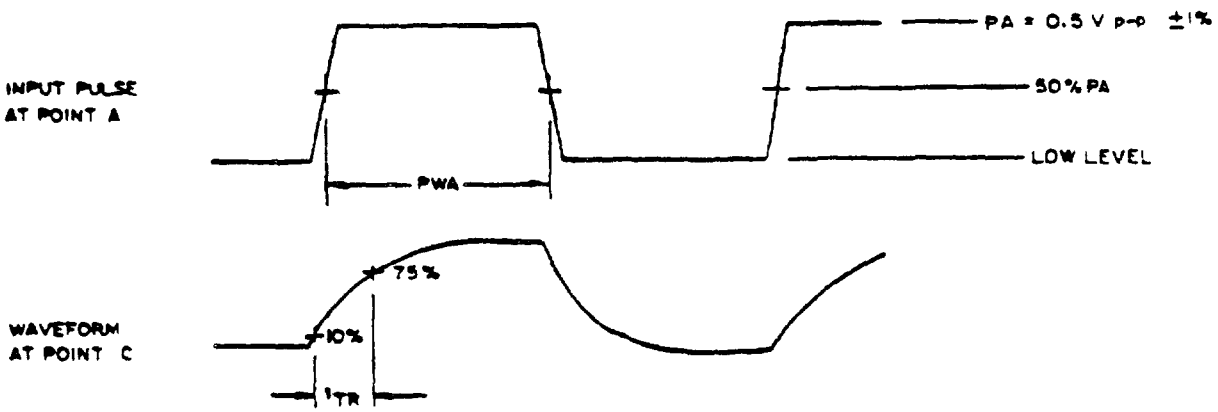
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NOTES
1 R1 = R2 = 51Ω ±5%, 1 W
2 R3 = 511Ω ±1%, 1/4 W

FIGURE 18 Power supply transient rise time test circuit for table XVII

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NOTES
1. Input pulse parameters:
   PWA = 10 ms ±10%.
   PRR = 50 Hz (reference).
   $t_{TLH} = t_{THL} = < 100$ ns (reference).
2. With point A open, adjust input as shown
3. With point A open, adjust V1 such that the low level of the input is 0.00 ±0.01 V.
4. Connect A to B, recheck the low level and pulse parameters at point A
5. Measure transient rise time (t_{TR}) at point C

FIGURE 19. Power supply transient rise time test circuit waveforms for table XVI.

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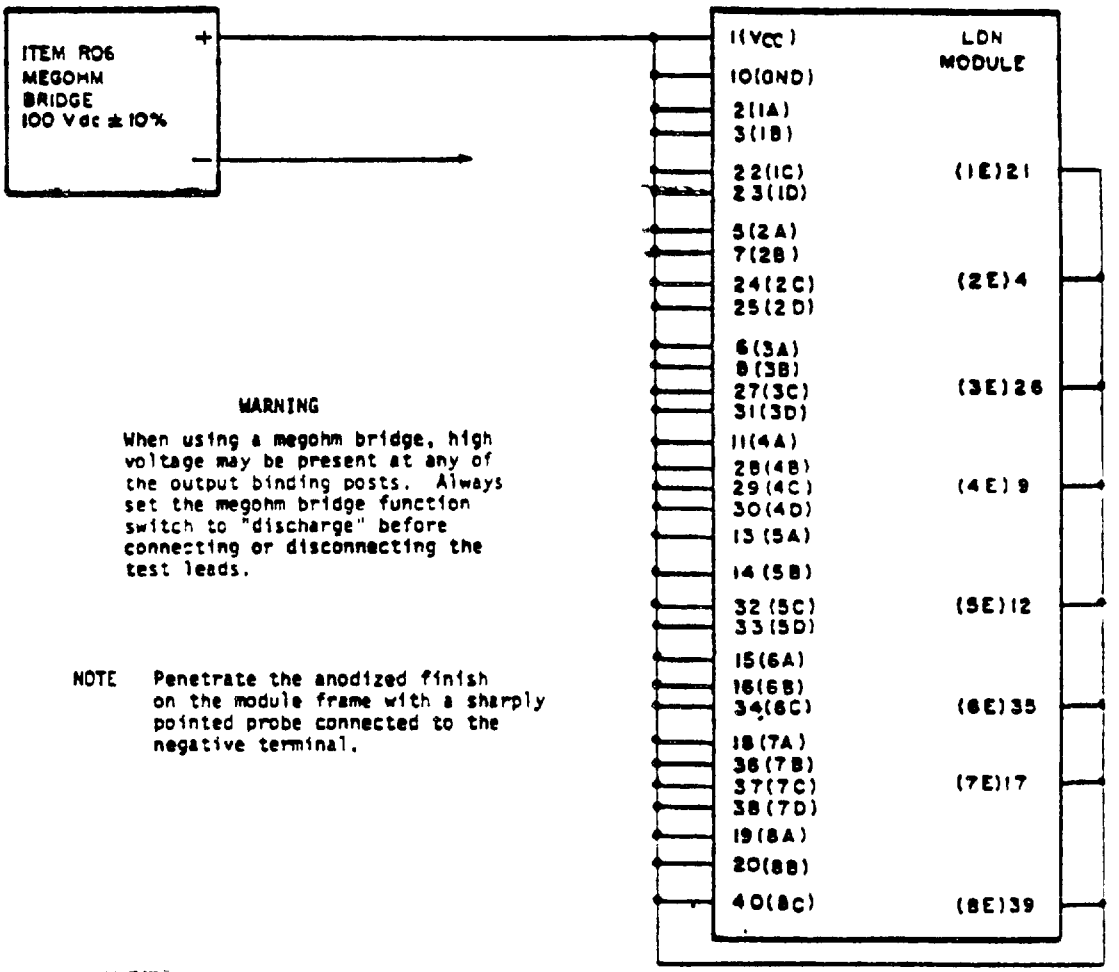


FIGURE 20. Isolation test circuit for table XVIII.

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TABLE XIX. Input capacitance test.

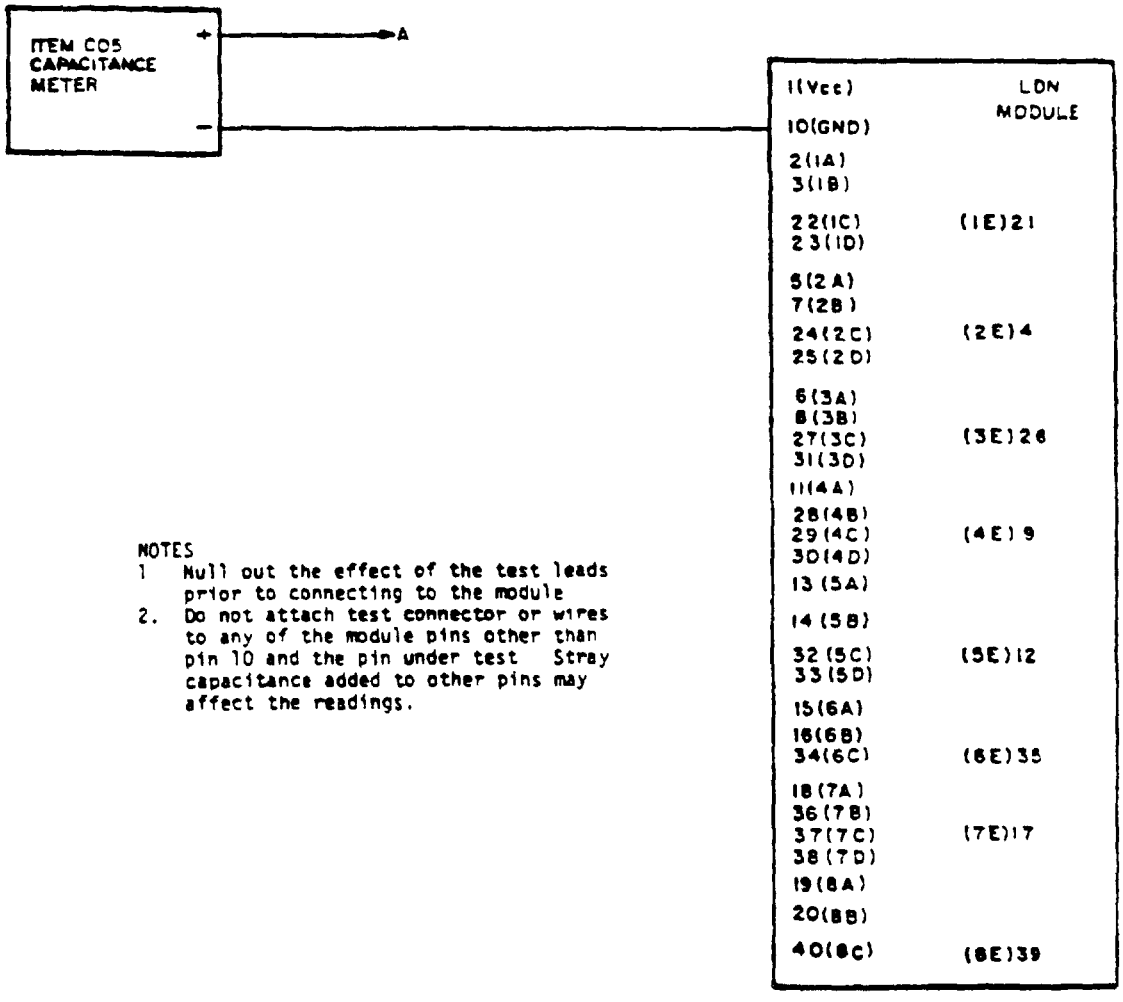
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
V1	B1 Input capacitance	3	2	15	pF	21
V2	A1 "	2	"	"	"	"
V3	C1 "	22	"	"	"	"
V4	D1 "	23	"	"	"	"
V5	A2 "	5	"	"	"	"
V6	B2 "	7	"	"	"	"
V7	C2 "	24	"	"	"	"
V8	D2 "	25	"	"	"	"
V9	A3 "	6	"	"	"	"
V10	C3 "	27	"	"	"	"
V11	B3 "	8	"	"	"	"
V12	D3 "	31	"	"	"	"
V13	A4 "	11	"	"	"	"
V14	B4 "	28	"	"	"	"
V15	C4 "	29	"	"	"	"
V16	D4 "	30	"	"	"	"
V17	A5 "	13	"	"	"	"
V18	B5 "	14	"	"	"	"
V19	C5 "	32	"	"	"	"
V20	D5 "	33	"	"	"	"

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TABLE XIX. Input capacitance test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
V21	A6 Input capacitance	15	2	15	pF	21
V22	B6 "	16	"	"	"	"
V23	C6 "	34	"	"	"	"
V24	A7 "	18	"	"	"	"
V25	B7 "	36	"	"	"	"
V26	C7 "	37	"	"	"	"
V27	D7 "	38	"	"	"	"
V28	B8 "	20	"	"	"	"
V29	A8 "	19	"	"	"	"
V30	C8 "	40	"	"	"	"

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NOTES
1. Null out the effect of the test leads prior to connecting to the module
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test Stray capacitance added to other pins may affect the readings.

FIGURE 21 Input capacitance test circuit for table XIX

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This amendment forms a part of MIL-M-28787/19B(EC), dated 21 April, 1983, and is approved for use within the Space and Warfare Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 4

TABLE I, Test conditions J1 and J2, Parameter column: Delete the word "circuit" (two places) and substitute "current" (two places).

PAGE 6

4.4.2, delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. Test conditions A, B, C, H, and U are stress tests and will be performed only at 25°C ±5°C. If test conditions A, B, C, H, or U are part of a sequence of tests, the order of testing shall be as follows: U, A, B, C, H where applicable, followed by the remaining tests in the sequence."

4.5.1, lines five and six Delete "C through R and T" and substitute "D through G, J through R, and T".

4.5.3, lines three and four Delete "C through R and T" and substitute "D through G, J through R, and T".

4.5.3, line six Delete "C through V" and substitute "D through G, J through T, and V".

PAGE 7

6.5, Output drive current, High level Delete "1 mA" and substitute "400 µA".

PAGE 24

FIGURE 9, NOTES Delete "Note 1" and substitute new Note 1

"1. R1 = 6.04 kΩ ±1%, 1/4 W, metal film."

PAGE 38

FIGURE 14, ITEM P01-2: Delete "3.5 ±1%" and substitute "3.5 V ±1%".

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